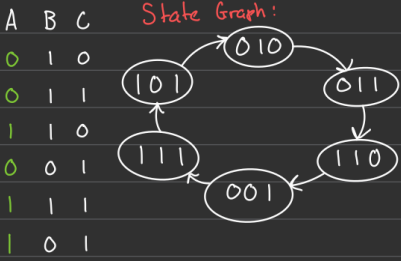


Quiz 3 Review

12.6 Design a circuit using D flip-flops that will generate the sequence 0, 0, 1, 0, 1, 1, 0, 1, 1 and repeat. Do this by designing a counter for any sequence of states such that the first flip-flop takes on this sequence. There are many correct answers, but do not duplicate states, because each state can have only one next state.

A must be given sequence randomly select unique values for B and C w/ A and A'



0→0→1→0→1→1
6 states → 3 FFs
1 FF = 2 states
States / 2 = Req. # of FFs

Transition Table

A	B	C	A'	B'	C'	D _A	D _B	D _C
0	0	0	X	X	X	X	X	X
0	0	1	1	1	1	1	1	1
0	1	0	0	1	1	0	1	1
0	1	1	1	1	0	1	1	0
1	0	0	X	X	X	X	X	X
1	0	1	0	1	0	0	1	0
1	1	0	0	0	1	0	0	1
1	1	1	1	0	1	1	0	1

BC \ A	0	1
00	X	X
01	1	0
11	0	1
10	0	0

D_A(A,B,C) = A'C + BC

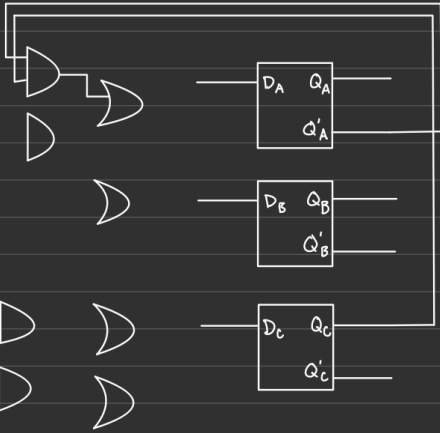
BC \ A	0	1
00	X	X
01	1	0
11	1	0
10	1	0

D_B(A,B,C) = A' + B'

BC \ A	0	1
00	X	X
01	1	0
11	0	1
10	1	1

D_C(A,B,C) = C' + A'B' + AB

Due to nature of D FF (Q' = D), D_{ABC} will be A', B', C'



D_A(A,B,C) = A'C + BC

D_B(A,B,C) = A' + B'

D_C(A,B,C) = C' + A'B' + AB

11.10 Convert by adding external gates:

a. a D flip-flop to a J-K flip-flop.

SHOW ANSWER

b. a T flip-flop to a D flip-flop.

SHOW ANSWER

c. a T flip-flop to a D flip-flop with clock enable.

SHOW ANSWER

a)

J	K	Q	Q'	D
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

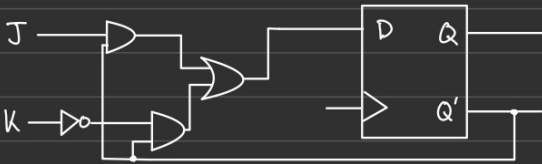
D type

Q' = D

KQ \ J	0	1
00	0	1
01	1	1
11	0	0
10	0	1

Q' = K'Q + JQ'

D = " "



When J = K = 1, the output toggles

b)

D	Q	Q'	T
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

T type

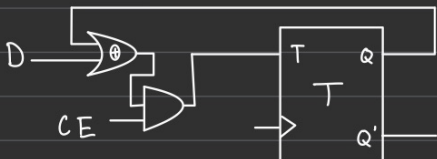
Q' = TQ' + T'Q

W/ toggle, if Q ≠ Q', then T must equal 1

D \ Q	0	1
0	0	1
1	1	0

T = DQ' + D'Q

T = D ⊕ Q

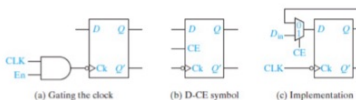


W/ Clock Enable

Q' = DCE + QCE'

D Flip-Flop with Clock Enable:

FIGURE 11-31 D Flip-Flop with Clock Enable © Cengage Learning 2014



When CE = 0, clock is disabled, Q' = Q
When CE = 1, clock is enabled, Q' = D
Characteristic equation: Q' = QCE + DCE
Note: CE and CE' in the two terms have MAX
Implement using a D FF and MAX
Q' = D = QCE' + DCE

Logan Bryars

ECE 3441 – Spring 2025

Classwork 6

2025/04/09

- Using T flip-flops, design a 3-bit counter which counts in the following sequence. Provide the state table, transition table, and next state equations for the T flip-flops.

Transition table:

A	B	C	A ⁺	B ⁺	C ⁺	T _A	T _B	T _C
0	0	0	0	0	1	0	0	1
0	0	1	0	1	1	1	0	0
0	1	0	1	1	0	2	1	0
0	1	1	0	1	0	3	0	1
1	0	0	0	0	0	4	1	0
1	0	1	1	0	0	5	0	1
1	1	0	1	1	1	6	0	1
1	1	1	1	0	1	7	0	1

$$Q^+ = T \oplus Q = TQ' + T'Q$$

When $Q \neq Q^+$, then T has to = 1

$$T_A f(A,B,C) = AB'C' + A'BC'$$

$$T_B'' = A'B'C + ABC$$

$$T_C'' = A'B'C' + AB'C + A'BC + ABC'$$

State table:

T_A

BC \ A	0	1
00	0	4
01	1	5
11	3	7
10	2	6

T_B

BC \ A	0	1
00		
01	1	
11		1
10		

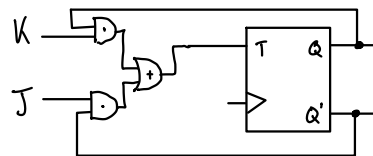
T_C

BC \ A	0	1
00	1	
01		1
11	1	
10		1

- Show how a T flip-flop can be converted to a J-K flip-flop by adding external gates.

J	K	Q	Q ⁺	T
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	0	1	1
1	1	1	0	1

$$T = KQ + JQ'$$



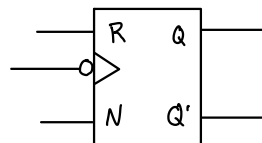
3. Suppose you are given a falling-edge triggered “R-N flip-flop” which behaves as follows:

If $R = 0$ and $N = 0$, then Q is toggled

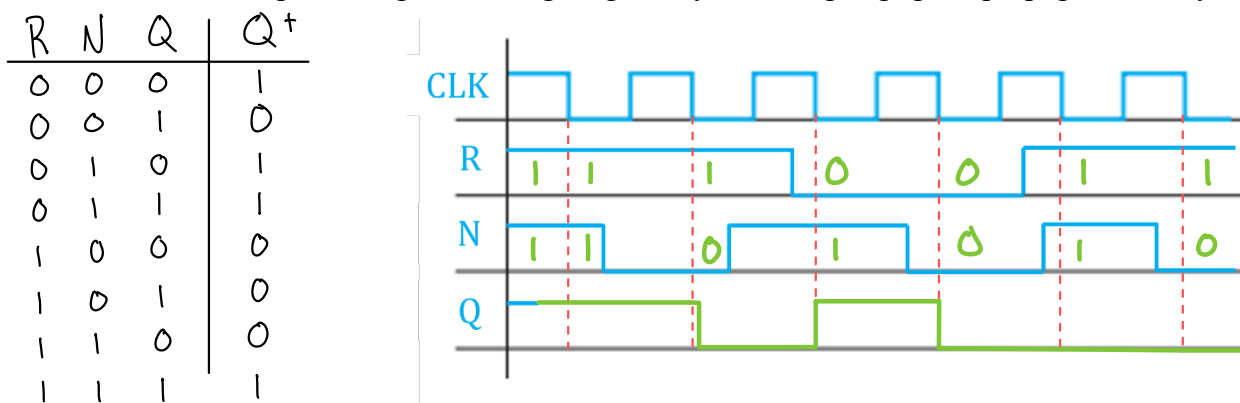
If $R = 0$ and $N = 1$, then Q is set to 1

If $R = 1$ and $N = 0$, then Q is set to 0

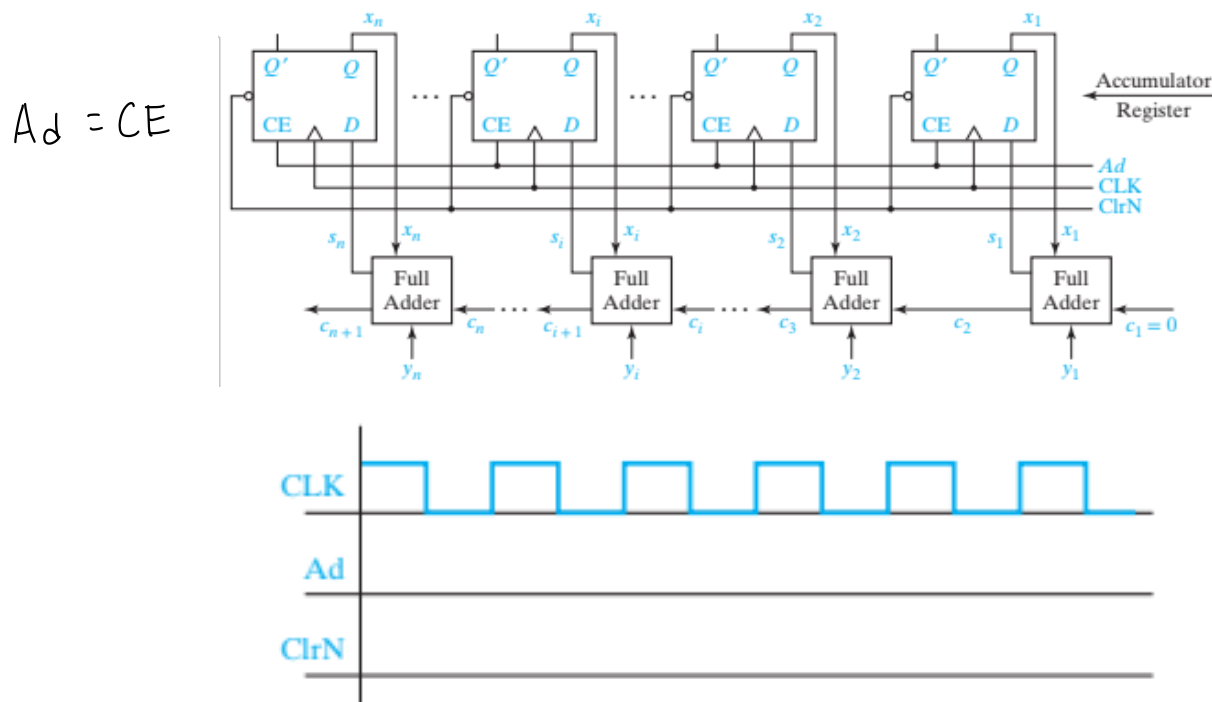
If $R = 1$ and $N = 1$, then Q is unaffected



Complete the given timing diagram by assuming negligible propagation delay.



4. Consider the 6-bit adder with an accumulator shown below. Suppose the X register contains a number from a previous calculation. We do not want this number. Instead, we want X to hold the value $3 \times Y$. ($X = X_5X_4X_3X_2X_1X_0$ and $Y = Y_5Y_4Y_3Y_2Y_1Y_0$). On the timing diagram, give values for Ad and $ClrN$ so that we will have $X = 3 \times Y$ held in the accumulator.



Classwork 6 Solutions

2025/04/09

1. Using T flip-flops, design a 3-bit counter which counts in the following sequence. Provide the state table, transition table, and next state equations for the T flip-flops.

001 → 011 → 010 → 110 → 111 → 101 → 100 → 000 → 001 → ...

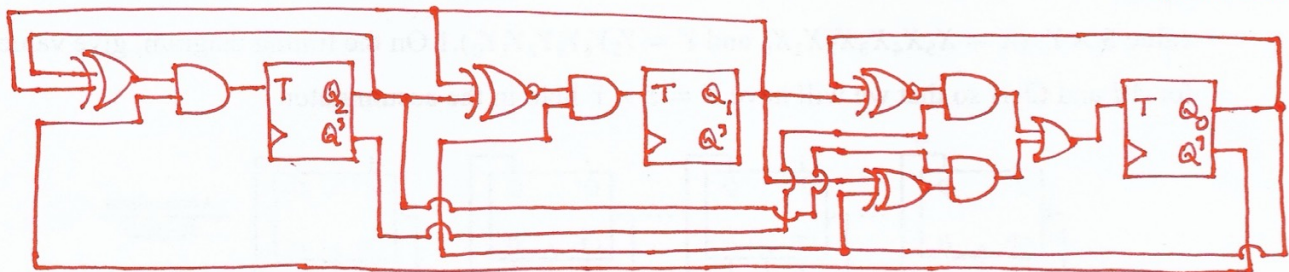
#	Present State	Next State	$Q_2 Q_1 Q_0$	$Q_2' Q_1' Q_0'$	$T_2 T_1 T_0$
1	001	011	0 0 1	0 0 1	0 0 1
2	011	010	0 0 1	0 1 1	0 1 0
3	010	110	0 1 0	1 1 0	1 0 0
4	110	111	0 1 1	0 1 0	0 0 1
5	111	101	1 0 0	0 0 0	1 0 0
6	101	100	1 0 1	1 0 0	0 0 1
7	100	000	1 1 0	1 1 1	0 0 1
8	000	001	1 1 1	1 0 1	0 1 0

$Q_2 Q_1 Q_0$	00	01	11	10
Q_2	0	0	0	1
Q_1	1	1	0	0

$Q_2 Q_1 Q_0$	00	01	11	10
Q_2	0	0	1	0
Q_1	1	0	0	1

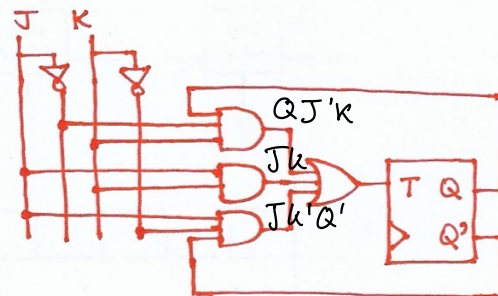
$Q_2 Q_1 Q_0$	00	01	11	10
Q_2	0	1	0	1
Q_1	1	0	1	0

$$\begin{aligned}
 T_2 &= Q_2' Q_1 Q_0' + Q_2 Q_1' Q_0' \\
 &= (Q_2' Q_1 + Q_2 Q_1') Q_0' \\
 &= (Q_2 \oplus Q_1) Q_0' \\
 T_1 &= Q_2' Q_1' Q_0 + Q_2 Q_1 Q_0 \\
 &= (Q_2' Q_1' + Q_2 Q_1) Q_0 \\
 &= (Q_2 \equiv Q_1) Q_0 \\
 T_0 &= Q_2' Q_1' Q_0' + Q_2' Q_1 Q_0 \\
 &\quad \dots + Q_2 Q_1' Q_0 + Q_2 Q_1 Q_0' \\
 &= Q_2' (Q_1 \equiv Q_0) + Q_2 (Q_1 \oplus Q_0)
 \end{aligned}$$



2. Show how a T flip-flop can be converted to a J-K flip-flop by adding external gates.

$$\begin{aligned}
 JK: \begin{cases} 00 : \text{no change} & \Rightarrow T = 0 \\ 01 : 0 & \Rightarrow T = Q \\ 10 : 1 & \Rightarrow T = Q' \\ 11 : \text{toggle} & \Rightarrow T = 1 \end{cases}
 \end{aligned}$$



3. Suppose you are given a falling-edge triggered "R-N flip-flop" which behaves as follows:

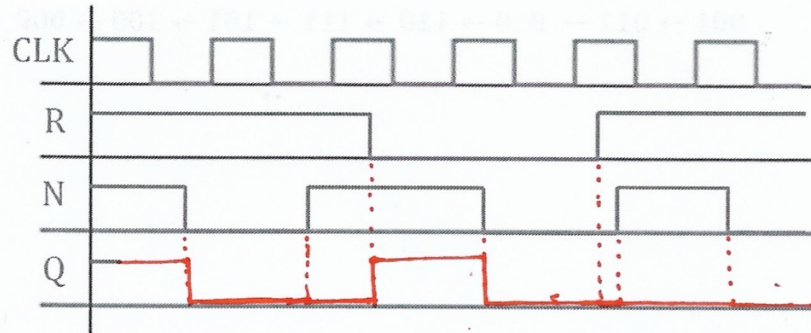
If $R = 0$ and $N = 0$, then Q is toggled

If $R = 0$ and $N = 1$, then Q is set to 1

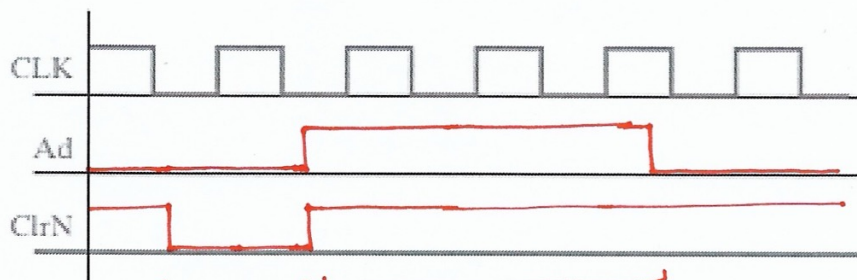
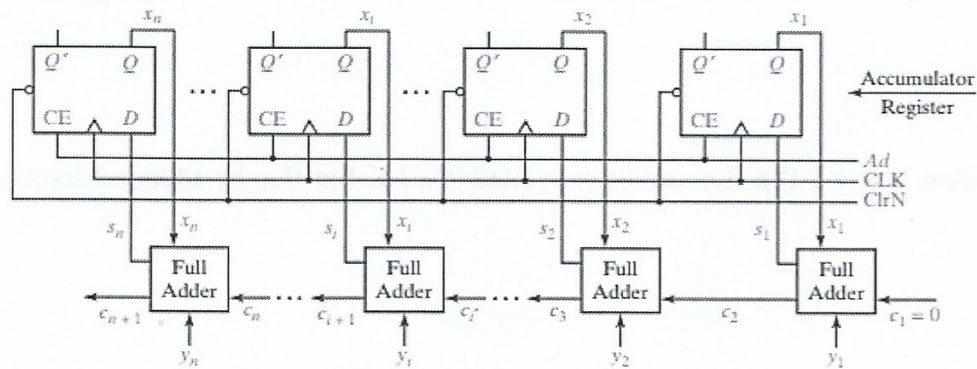
If $R = 1$ and $N = 0$, then Q is set to 0

If $R = 1$ and $N = 1$, then Q is unaffected

Complete the given timing diagram by assuming negligible propagation delay.



4. Consider the 6-bit adder with an accumulator shown below. Suppose the X register contains a number from a previous calculation. We do not want this number. Instead, we want X to hold the value $3 \times Y$. ($X = X_5X_4X_3X_2X_1X_0$ and $Y = Y_5Y_4Y_3Y_2Y_1Y_0$). On the timing diagram, give values for Ad and $ClrN$ so that we will have $X = 3 \times Y$ held in the accumulator.



this gets
rid of
value in X

Page 2

this causes X to
accumulate Y 3
times making it ~~3~~
 $3 \times Y$.

ECE 3441 Fall 2022

Quiz 3

VERSION B

- Closed book, closed notes
- No cheat sheet is allowed.
- Only four function calculators are allowed.
- Show all details of your work to obtain credit (The answer alone will not suffice).
- You may not talk during the quiz to others or to yourself.

Please do not open booklet unless instructed to do so, or substantial penalty will result.

First Name Last Name,

Peoplesoft ID,

Date *DD/MM/YYYY*

Question	Points	Maximum
1		40
2		30
3		40

Total: _____/100 possible points + 10 Bonus

Question #1:

An L-P flip-flop works as follows:

When $L = 0$, the next state of the flip flop remains the same.

When $LP = 11$, the next state of the flip-flop is 1.

When $LP = 10$, the next state of the flip-flop changes.

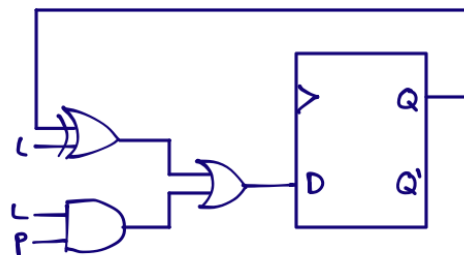
- Find the truth table for this flip-flop.
- From the truth table derive the characteristic equation for the flip-flop.
- Construct this L-P flip-flop from a D flip-flop using the minimum number of gate (3 gates)

L	P	Q	Q ⁺
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Q	LP	
	0	1
00	0	1
01	0	1
11	1	1
10	1	0

$$Q^+ = L'Q + LQ' + LP$$

$$Q^+ = (L \oplus Q) + LP$$



Question #1:

An L-P flip-flop works as follows:

When $L = 0$, the next state of the flip flop remains the same.

When $LP = 11$, the next state of the flip-flop is 1.

When $LP = 10$, the next state of the flip-flop changes. *Toggles*

- Find the truth table for this flip-flop.
- From the truth table derive the characteristic equation for the flip-flop.
- Construct this L-P flip-flop from a D flip-flop using the minimum number of gate (3 gates)

a)

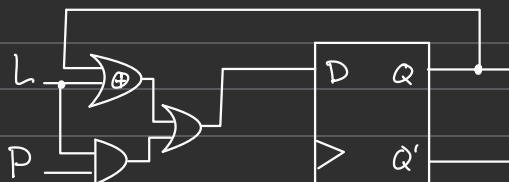
L	P	Q	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

b)

$Q^+ f(L, P, Q)$

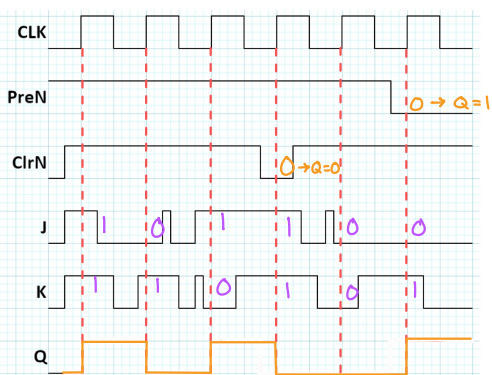
PQ \ L	0	1
	00 01 11 10	04 5 7 6

$Q^+ = L'Q + LP$
 $= (L \oplus Q)$

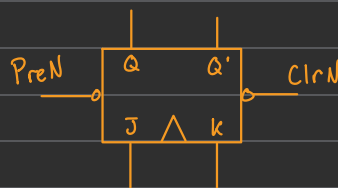


Question #2:

Complete the image for a rising-edge triggered J-K flip-flop. PreN and ClrN are asynchronous inputs. Assume that propagation delay is negligible. Note that the first part of Q is drawn for you.



$$\begin{aligned} \text{Pre}N &= S' \\ \text{Clr}N &= R' \end{aligned}$$

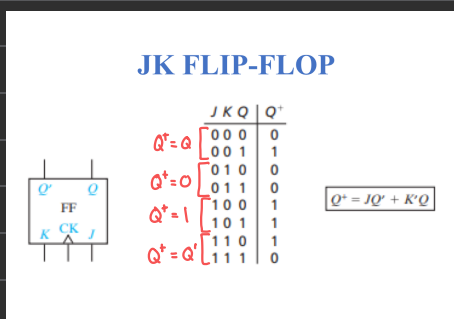


PreN = 1 $\rightarrow$ no change

$ClrN=1 \rightarrow$ no change

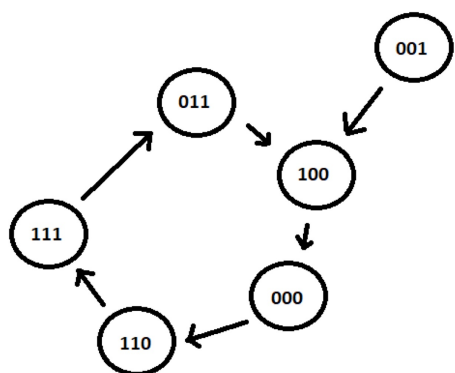
Set $P_{reN} = 0 \rightarrow Q^+ = 1$

Reset $C_{lr} N = 0 \rightarrow Q^+ = 0$



Question #3:

Implement the following counter using T flip-flops and draw its associated circuit.



A B C	A ⁺ B ⁺ C ⁺	A B C	A ⁺ B ⁺ C ⁺	T _A	T _B	T _C
0 0 1	1 0 0	0 0 0	1 1 0	1	1	0
1 0 0	0 0 0	0 0 1	1 0 0	1	0	1
0 0 0	1 1 0	0 1 0	X X X	X	X	X
1 1 0	1 1 1	0 1 1	1 0 0	1	1	1
1 1 1	0 1 1	1 0 0	0 0 0	1	0	0
0 1 1	1 0 0	1 0 1	X X X	X	X	X
		1 1 0	1 1 1	0	0	1
		1 1 1	0 1 1	1	0	0

BC \ A	0	1
00		
01		
11		
10		

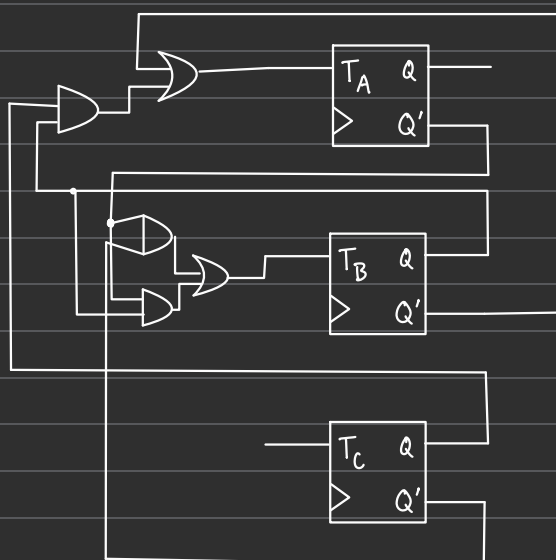
$$T_A = B' + BC$$

BC \ A	0	1
00	1	
01		X
11	1	
10	X	

$$T_B = A'C' + A'B$$

		A	
		0	1
BC	00		
	01	1	X
	11	1	
	10	X	1

$$T_c = A'C + BC'$$



11.27a. Construct a D flip-flop using an inverter and an S-R flip-flop.

b. If the propagation delay and setup time of the S-R flip-flop in (a) are 2.5 ns and 1.5 ns, respectively, and if the inverter has a propagation delay of 1 ns, what are the propagation delay and setup time of the D flip-flop of part (a)?

Excitation Table:

Q	Q ⁺	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

S	R	Q	Q ⁺
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

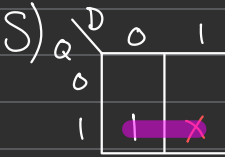
D	Q	Q ⁺
0	0	0
0	1	0
1	0	1
1	1	1

S-R:

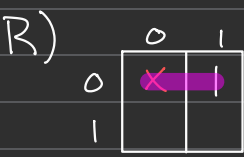
$$Q^+ = S + R'Q$$

D:

$$Q^+ = D$$



$$S = Q^+$$
$$Q^+ = D$$

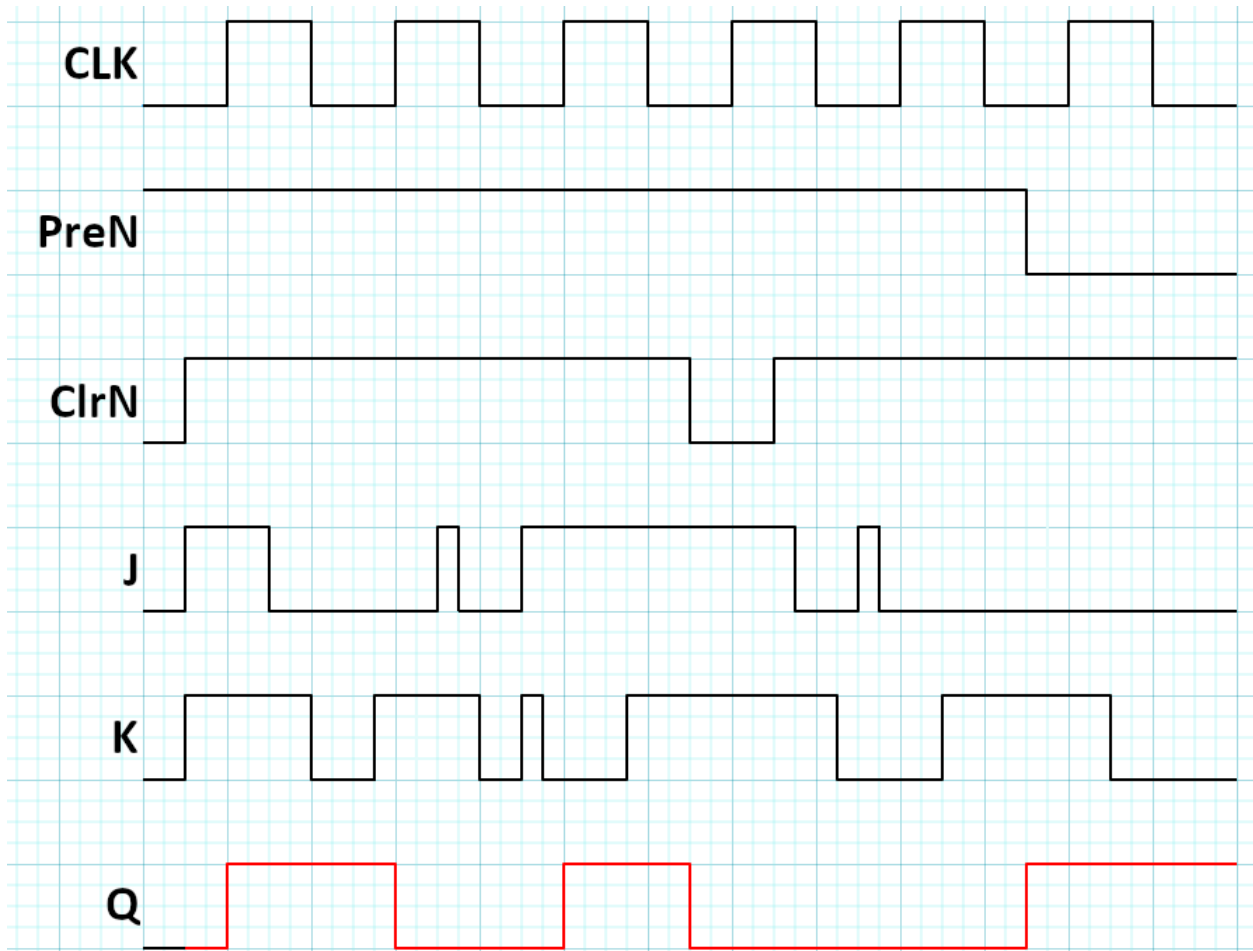


$$R = Q^{+'}$$
$$Q^{+'} = D'$$



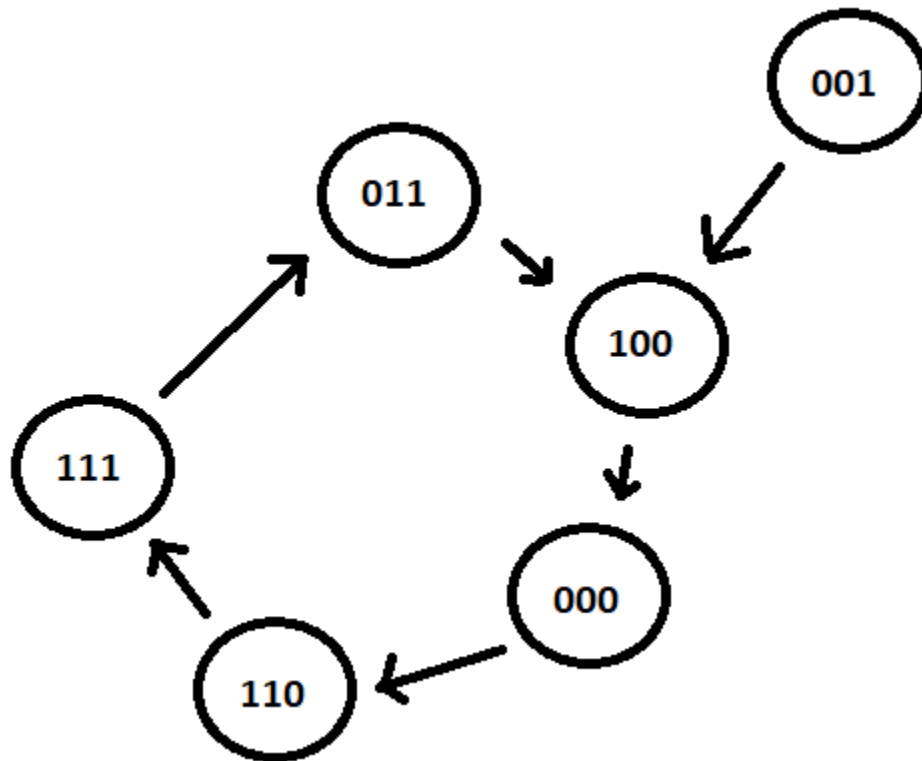
Question #2:

Complete the image for a rising-edge triggered J-K flip-flop. PreN and ClrN are asynchronous inputs. Assume that propagation delay is negligible. Note that the first part of Q is drawn for you.



Question #3:

Implement the following counter using T flip-flops and draw its associated circuit.



A	B	C	A+	B+	C+	TA	TB	TC
0	0	0	1	1	0	1	1	0
0	0	1	1	0	0	1	0	1
0	1	0	X	X	X	X	X	X
0	1	1	1	0	0	1	1	1
1	0	0	0	0	0	1	0	0
1	0	1	X	X	X	X	X	X
1	1	0	1	1	1	0	0	1
1	1	1	0	1	1	1	0	0

TA	A	
BC	0	1
00	1	1
01	1	X
11	1	1
10	X	0

TB	A	
BC	0	1
00	1	0
01	0	X
11	1	0
10	X	0

TC	A	
BC	0	1
00	0	0
01	1	X
11	1	0
10	X	1

TA	A	
BC	0	1
00	1	1
01	1	X
11	1	1
10	X	0

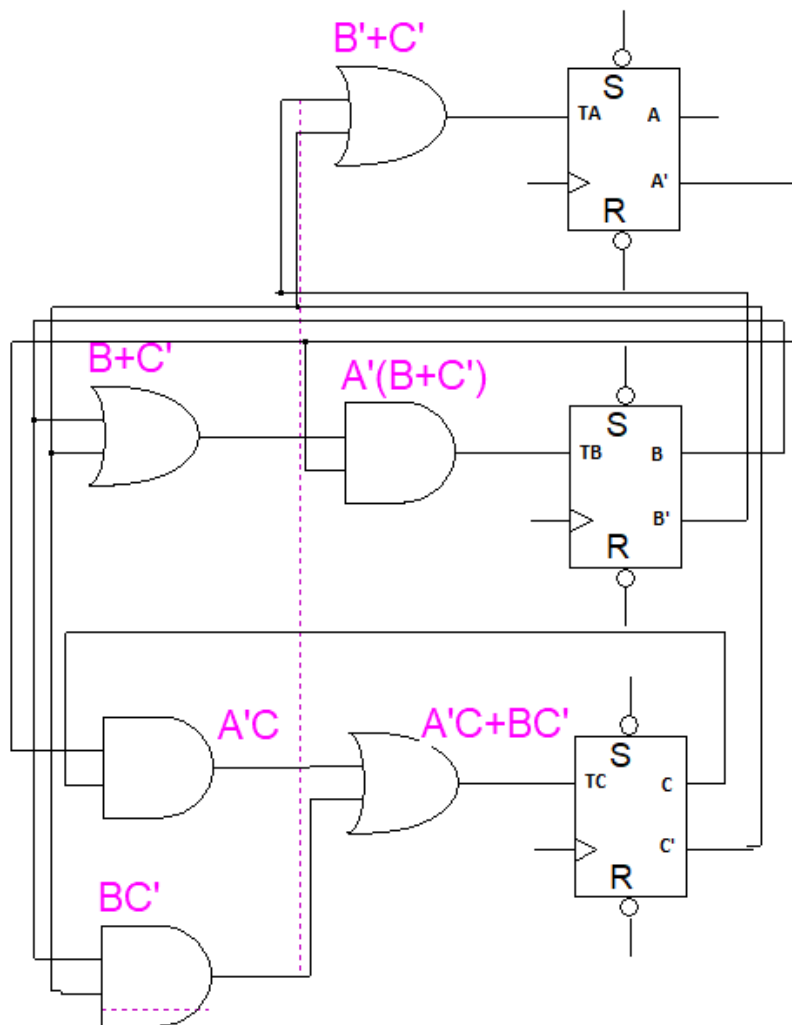
TB	A	
BC	0	1
00	1	0
01	0	X
11	1	0
10	X	0

TC	A	
BC	0	1
00	0	0
01	1	X
11	1	0
10	X	1

$$TA = B' + C$$

$$TB = A'B + A'C'$$

$$TC = A'C + BC'$$



T FLIP-FLOP



T	Q	Q^+
0	0	0
0	1	1
1	0	1
1	1	0

$$Q^+ = T'Q + TQ' = T \oplus Q$$

JK FLIP-FLOP



J	K	Q	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

$$Q^+ = JQ' + K'Q$$

SR FLIP-FLOP



S	R	Q	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	-
1	1	1	-

inputs not allowed

$$Q^+ = S + R'Q \quad (SR = 0)$$

D FLIP-FLOP



D	Q	Q^+
0	0	0
0	1	0
1	0	1
1	1	1

$$Q^+ = D$$